

ELECTRICITY

Types of charges:

- positive
- Negative

A charge is a qty of $Q = n e$ ($1.6 \times 10^{-19} \text{ C}$)

$$Q = n e$$

Electric forces and electric fields:

Electroly refers to the flow of ^{electrons} electric charges.

Elect (study of $\pm$ charges in motion)

Electrostatic study of electric charges at rest.

The electrons are not found in the nucleus bcz of their high speed.

Conservation of charge:

The net charge in an isolation system remains constant

- Conductors: electrons move easily $\propto 4$
- Insulators: electrons don't move. $\propto 4$
- Semi conductors: In btm. $\propto 4$

Valence electrons are those electrons which are located in the outermost shell of an atom

The nbr of valence e^- determines the electronic properties of that atom

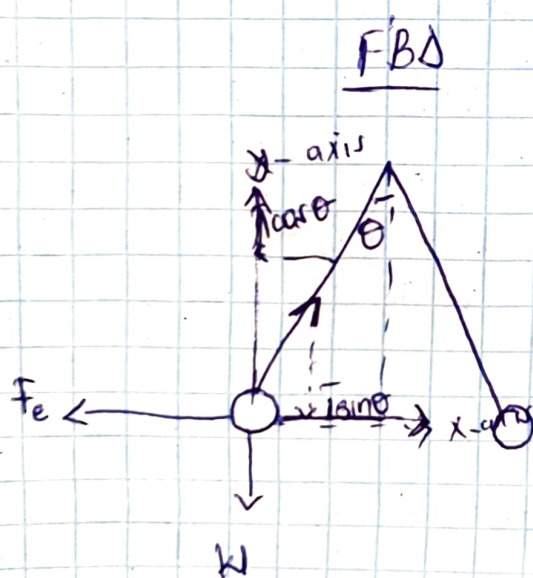
< 4 : Conductor

> 4 : Insulator

$= 4$: Semi-conductor

3 ways that objects can be given a net charge:

- charging by friction: mainly charging insulators
- charging by contact: mainly metals & other conductors
- charging by induction: Metals & other conductors. It doesn't make contact.



60 it

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

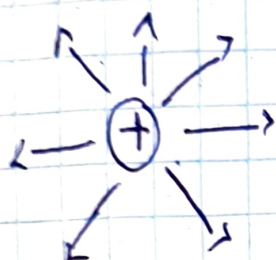
x-axis	y-axis
$-F_e$ $T \sin \theta$	$T \cos \theta$ W
$\Sigma x: -F_e + T \sin \theta$	$\Sigma y: T \cos \theta + W$

$$\text{Net } T = \sqrt{(-F_e + T \sin \theta)^2 + (T \cos \theta + W)^2}$$

Electric field: It is a region around a charged body within which it can exert its electrostatic force.

Electric field Intensity: $\vec{E} = k \frac{Q}{r^2}$
 $\downarrow$
 Vector only

Electric field lines:



E. Flux: nbr of lines of field passing through the cross section area.

nbr of ~~flux~~ lines: $\frac{\text{Electric field}}{\text{Area}}$

$$\Phi = \vec{E} \cdot \vec{A} \cos \theta$$

If not parallel/perpendicular to the field

Force on a charged particle.

$$a = \frac{F_e}{m} = \frac{q E}{m}$$

$$V = \sqrt{2 |q y|}$$

$$\Phi = \frac{q}{\epsilon_0} \rightarrow \text{charge}$$

$\uparrow$

Gauss's law

$$\Phi_E = \vec{E} \cdot \Delta \vec{A}$$

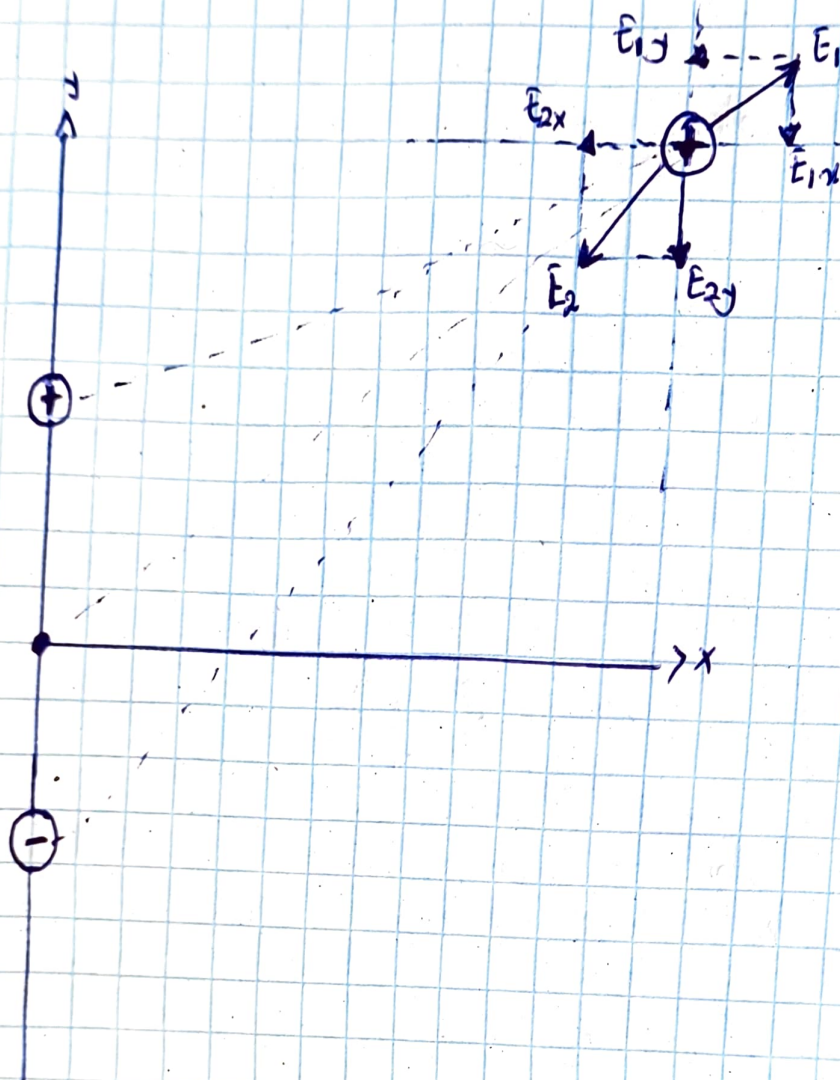
$$F = m \vec{a}$$

$$a = \frac{F_e}{m} = \frac{q \cdot E}{m}$$

$$F = q \cdot \vec{E}$$

$$V_y = \sqrt{2 |q y|} = \sqrt{\frac{2 q q E y}{m}}$$

$$K = \frac{1}{2} m v_g^2 = q E_y y$$



$$F = \frac{k q_1 q_2}{r^2}$$

$$F_e = \frac{9 \times 10^9 \times 1.6 \times 10^{-19} \times -1.6 \times 10^{-19}}{(5.3 \times 10^{-11})^2}$$

$$F_e = 8.2 \times 10^{-8} \text{ N}$$

$$F_g = \frac{6.67 \times 10^{-11} \times 1.67 \times 10^{-27} \times 9.1 \times 10^{-31}}{(5.3 \times 10^{-11})^2} \times 6.67 \times 10^{-11}$$

$$F_g = 3.6 \times 10^{-47} \text{ N}$$

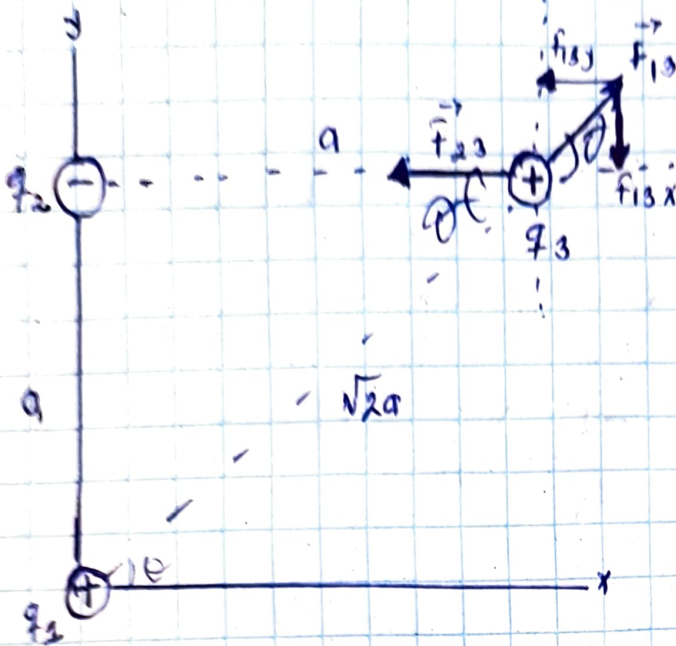
Eg 2:

2A

$$q_1 = q_3 = 5 \mu C$$

$$q_2 = -2 \mu C$$

$$a = 0.1 m$$



$$\phi = \frac{Q}{H}$$

x	y
$\Sigma x = -1.045$	$\Sigma y = 7.95$

$$|\vec{F}| = \sqrt{(-1.045)^2 + (7.95)^2} = 8.023 N$$

$$\begin{aligned} F_{23} &= \left| \frac{k q_2 q_3}{r^2} \right| \\ &= \frac{9 \times 10^9 \times -2 \times 10^{-6} \times 5 \times 10^{-6}}{(0.1)^2} \\ &= 9 N \end{aligned}$$

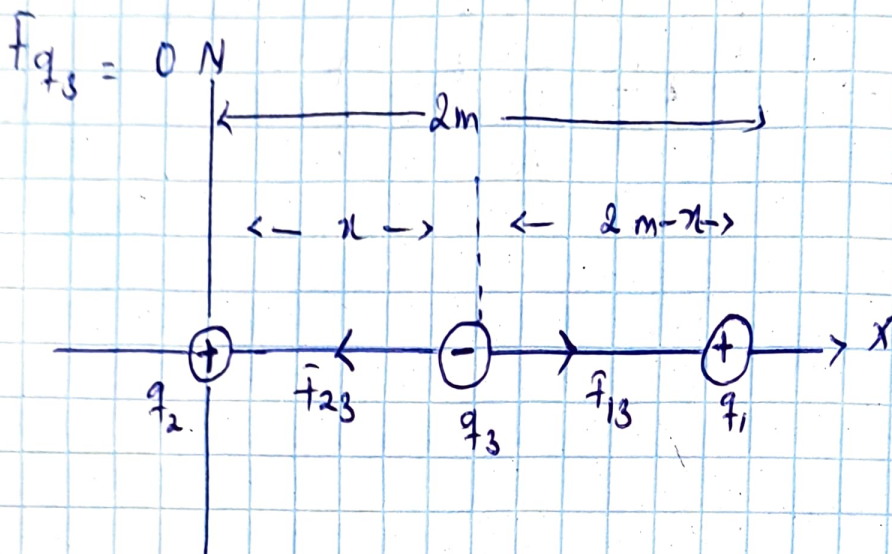
$$\begin{aligned} F_{13} &= \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 5 \times 10^{-6}}{(\sqrt{2} \times 0.1)^2} \\ &= 11.25 N \end{aligned}$$

3

$$q_1 = 15.0 \mu\text{C}$$

$$u = 2 \text{ m}$$

$$q_2 = 6 \mu\text{C}$$



$$F_3 = F_{13} - F_{23}$$

$$F_{13} = F_{23}$$

$$F_3 = 0$$

$$\frac{9 \times 10^9 \times 15 \times 10^{-6} \times q_3}{(2-x)^2} = \frac{9 \times 10^9 \times 6 \times 10^{-6} \times q_3}{x^2}$$

$$(15 \times 10^{-6}) x^2 = (2-x)^2 (6 \times 10^{-6})$$

$$(15 \times 10^{-6}) x^2 = 4 - 4x + x^2$$

$$2.5 x^2 = (2-x)(2-x)$$

$$2.5 x^2 = 4 - 2x - 2x + x^2$$

$$2.5 x^2 = 4 - 4x + x^2$$

$$1.5 x^2 + 4x - 4 = 0$$

$$x_1 = 0.774 \text{ m}$$

$$m = 3 \times 10^{-12} \text{ kg}$$

$$E = 8 \times 10^3 \text{ N/C}$$

$$E = \frac{F}{q}$$

$$E q = F_g$$

$$6 \times 10^{-12} \times q = 3$$

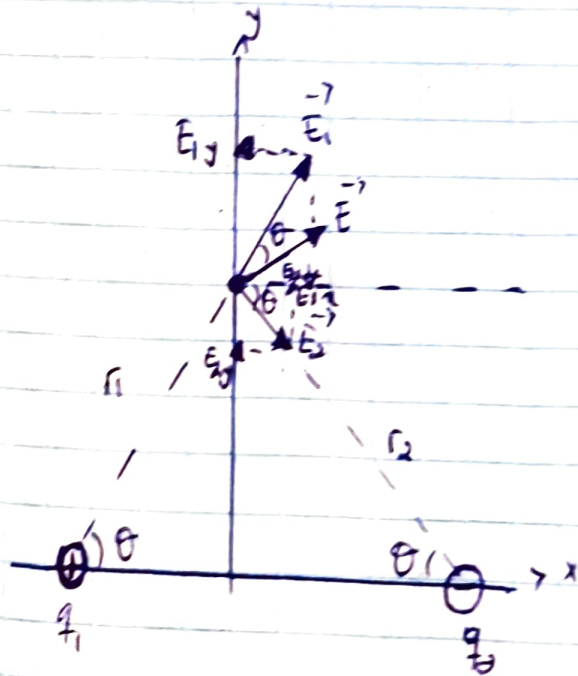
$$-6 \times 10^3 \times q = 3 \times 10^{-12} \times q = 3$$

$$q = -4.9 \times 10^{-16} \text{ C}$$



5.

$$q) E = \frac{F}{q} = \frac{k q_1}{r^2}$$



x

y

$$E_1 \cos \theta$$

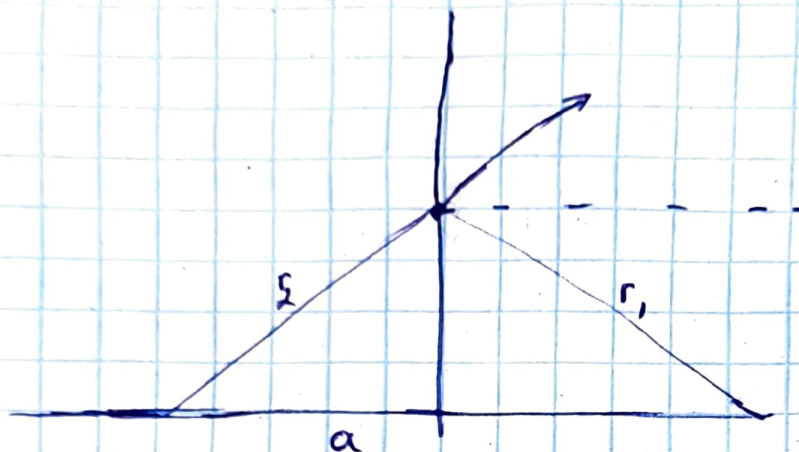
$$E_2 \cos \theta$$

$$E_1 \sin \theta$$

$$- E_2 \sin \theta$$

$$\Sigma x = 0$$

(B)



$$E_1 = E_2$$

x	y
$E_2 \cos \theta + E_2 \cos \theta$ $2E_2 \cos \theta$	$E_1 \sin \theta - E_2 \sin \theta$ $= 0$

ELECTRICAL POTENTIAL

$$u = k Q_1 Q_2 \int \frac{dr}{r^2}$$

$$\Rightarrow -k Q_1 Q_2 \int r^{-2} dr$$

$$= -k \frac{Q_1 Q_2}{r^{-1}}$$

$$u = \frac{k Q_1 Q_2}{r} \text{ E.P. Energy}$$

$$-\frac{\partial u}{\partial r} = \frac{k Q_1 Q_2}{r^2}$$

$$\partial u = -\frac{k Q_1 Q_2}{r^2} dr$$

$$u = -k Q_1 Q_2 \int \frac{dr}{r^2}$$

$$E = \frac{kQ}{r^2}$$

$$V = \frac{kQ}{r}$$

$$V \cdot Q = \frac{kQ}{r}$$

$$U = V \cdot Q$$

Potential difference:

$$V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{l}$$

Motion of a particle in a uniform electric field.

$$F = qE = ma$$

$$a = \frac{qE}{m}$$

$$= \frac{eE}{m}$$

If a particle has a +ve charge its acceleration is in the direction of the $\vec{E}$ & if it has a -ve charge, its direction is opposite to the direction of $\vec{E}$ field lines.

for speed:

$$V^2 = u^2 + 2a(d - d_0)$$

$$V^2 = 2ad$$

$$V = \sqrt{2ad}$$

$$V = \sqrt{2\left(\frac{qE}{m}\right)d}$$

$$V = \sqrt{\frac{2qEd}{m}}$$

Since $a = \frac{qE}{m}$

speed of a particle

time: Since the electric force acts in the vertical direction then the horizontal motion is treated as if it is under constant velocity so,

$$x_f = x_i + v_{xi} t$$

$$t = \frac{x_f - x_i}{v_{xi}}$$

t : time to leave the field

x_f : length of the plate.

Vertical position (y) as it leaves the field;

Since in the vertical direction there's a constant force, means there's also a constant acceleration so,

$$y = y_0 + v_{yi} t + \frac{1}{2} a t^2$$

Electric field $\vec{E}$ at some point is defined as the electric force $\vec{F}$ acting on a small test charge placed at that point divided by the magnitude q_0 of test charge.

$$\vec{E} = \frac{\vec{F}_e}{q_0}$$

$\vec{E}$ due to a continuous charge distribution is

$$\vec{E} = k_e \int \frac{dq}{r^2} \hat{r}$$

Recall on Gauss's law:

$$\phi = \oint \vec{E} \cdot d\vec{A} = E \oint dA = E \cdot A = E \cdot 4\pi r^2$$

Since it's a sphere $A = 4\pi r^2$

Since $E = \frac{kq}{r^2}$

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$$\phi = \frac{kq}{r^2} \cdot 4\pi r^2$$

$$\phi = kq 4\pi$$

Since $k = \frac{1}{4\pi\epsilon_0}$

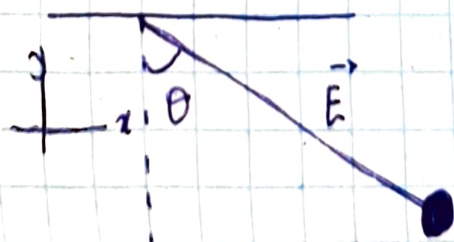
$$\phi = \frac{q 4\pi}{4\pi\epsilon_0}$$

$\phi = \frac{q}{\epsilon_0}$ which is Gauss's law

- The net electric flux through a closed surface that surrounds no charge is zero.
- The net flux through any closed surface surrounding a point charge q is given by q/ϵ_0 and is independent of the shape of that surface.

Ex:

$m = 1g$ $\vec{E} = (3\hat{i} + 5\hat{j}) \times 10^5 \text{ N/C}$ $\theta = 37.0^\circ$



Gauss's law states that the net flux of electric field over the gaussian surface

$$V = \vec{E} \times d$$

$$V = \frac{kq}{r}$$

$$kQ_1 Q_2 \int \frac{1}{r^2} dr = kQ_1 Q_2 \left[\frac{r^{-1}}{-1} \right] = - \frac{kQ_1 Q_2}{r}$$

Electric potential Energy : $\frac{kQ_1 Q_2}{r} = \text{J}$

$$V = \frac{\text{Electric Potential Energy}}{q} \quad (\text{V})$$

$$W = V \cdot q$$

Capacitors:

They store electric charges. Ability to store charges is called Capacitance.

$$C = \frac{Q}{V}$$

from $Q = CV$

SI is Farad
It depends on r

$\Rightarrow$

$$C = \frac{Q}{\frac{1}{4\pi\epsilon_0} \frac{Q}{R}} = 4\pi\epsilon_0 R$$

$$\Delta V = \frac{Qd}{\epsilon_0 A}$$

$\Rightarrow$

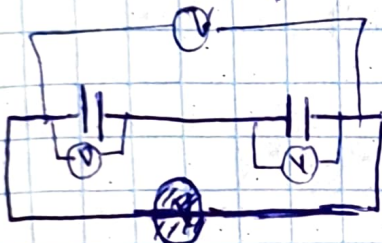
$$C = \frac{Q}{\Delta V} = \frac{\epsilon_0 A}{d}$$

$$C_1 = \frac{Q_1}{V}$$

$$C_2 = \frac{Q_2}{V}$$

$$C_1 + C_2 = \frac{Q_1 + Q_2}{V}$$

$$TC = \frac{Q_1 + Q_2}{V}$$



In series $Q_1 = Q_2$

$$C = \frac{Q}{V}$$

$$V = V_1 + V_2$$

$$C_1 + C_2 = \frac{Q_1}{V_1} + \frac{Q_2}{V_2}$$

$$V = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$TC = \frac{V_2 Q_1 + Q_2 V_1}{V_1 + V_2}$$

$$V = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

$$TC = V_2 Q + V_1 Q$$

$$\frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$TC = \frac{Q(V_1 + V_2)}{V_1 V_2}$$

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$$

ENERGY IN CAPACITORS

$$U = \frac{1}{2} VQ = \frac{1}{2} \frac{Q}{C} \cdot Q$$

$$= \frac{1}{2} \frac{Q^2}{C}$$

ELECTRIC CURRENT

It is the measure of the rate of flow of electric charge through a given cross-section of area.

$$I = Q/t$$

RESISTANCE

$$R = \frac{\rho L}{A}$$

ρ : Resistivity
 L : length

A : Cross section area.

It depends on length, temperature

Resistivity depends on the surface

OHM'S LAWS

$$V = IR$$

$$I = V/R$$

$$R = V/I$$

The voltage is directly proportional to the product of current and Resistance.

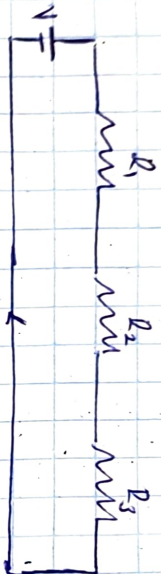
EMF is the voltage of a cell when a circuit is open. Max possible voltage of a battery.

$$\mathcal{E} = V + Ir$$

r : Internal resistance
 V : terminal voltage

Voltage across terminals, voltage lost inside battery

SERIES CIRCUIT



$$V = IR$$

$$V = IR$$

$$V = IR_1 + IR_2 + IR_3$$

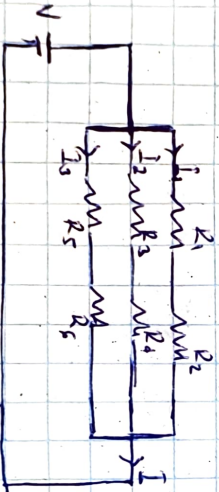
$$V = I (R_1 + R_2 + R_3)$$

$$\frac{V}{I} = R_1 + R_2 + R_3$$

$$R_T = R_1 + R_2 + R_3$$

$$V = V_1 + V_2 + \dots$$

PARALLEL CIRCUITS



Voltage is the same.

$$I_T = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} + \dots$$

$$\frac{V}{R} = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

Magnetism is a class of physical phenomena that are mediated by magnetic fields.

Magnetic field is any region around a magnet. $\vec{B}$

$$\vec{F} = q \vec{B}$$

$$F = k B$$

$$F = k q \quad \vec{F} = k B \vec{v}$$

$$\vec{F} = q \vec{v} \times \vec{B}$$

$$F_{\text{mag}} = q |\vec{v}| |\vec{B}| \sin \theta = q v B \sin \theta$$

θ : angle b/w the $\vec{v}$ & Magnetic field vector

When a charged particle is put in a magnetic field, the particle is going to revolve in circular motion.

$$F_c = F_{\text{magnetic}}$$

$$\Leftrightarrow \frac{m v^2}{r} = |q| v B$$

$$\frac{m v^2}{r} = |q| v B$$

$$\frac{m v^2}{r} = |q| v B$$

$$r = \frac{m v}{|q| B}$$

When
field,
current

Motion of a charged particle in

$$w = \frac{v}{r} = \frac{191 \text{ s}}{m}$$

$$f = \frac{2\pi r}{v} = \frac{2\pi}{w} = \frac{2\pi m}{191 \text{ s}}$$

When there's an electric field:

$$\begin{aligned} F_{\text{tot}} &= F_{\text{electric}} + F_{\text{mag}} = qE + qv \times B \\ &= q(\vec{E} + \vec{v} \times \vec{B}) \end{aligned}$$

THE HALL EFFECT

When a current carrying conductor is placed in a magnetic field, a p.d is generated in a direction $\perp$ to both current & magnetic field.